



C#

Level 1

Array Operations Min Max



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Definition

LINQ Array Operations

-  System.Linq provides useful methods for working with arrays and collections.
-  Min() finds the smallest value, while Max() finds the largest.

Explanation

 Import System.Linq

using System.Linq;

 This is required to use LINQ extension methods such as *Min()* and *Max()*.

 *Min()*

 Returns the smallest element in the array.

 *Max()*

 Returns the largest element in the array.

💡 Example

```
int[] numbers = { 51, -1, 2, 14, 18, 40, 178 };  
Console.Write("Array element: { ");  
foreach (int number in numbers)  
{  
    Console.Write(number + ", ");  
}  
Console.WriteLine("}");  
Console.WriteLine("Smallest Element: " + numbers.Min());  
Console.WriteLine("Largest Element: " + numbers.Max());
```

➡ Result

```
Array element: { 51, -1, 2, 14, 18, 40, 178, }  
Smallest Element: -1  
Largest Element: 178
```

Note

- ◆ `Min()` and `Max()` are LINQ methods.
- ◆ Other useful LINQ methods include `Sum()`, `Average()`, `Count()`, `First()`, and `Last()`.
- ◆ For this lesson, focus on `Min()` and `Max()`.

Comparison

Min()

- ✓ Finds the smallest value.
- ✓ Can replace a manual minimum-search loop.

Max()

- ✓ Finds the largest value.
- ✓ Can replace a manual maximum-search loop.

✦ Summary

- ✓ using `System.Linq`; enables LINQ extension methods.
- ✓ `Min()` → smallest value.
- ✓ `Max()` → largest value.
- ✓ Both can be used directly with an array.

Thank you 😊